

POWER BI · DATA ANALYTICS PORTFOLIO

E-Commerce Sales Analytics Dashboard Documentation

A complete summary of the dataset, dashboard design, interactive features, and business insights behind the project.



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1 Dataset Summary

This project analyzes an e-commerce sales dataset sourced from Kaggle — Ecommerce sales dataset. The dashboard provides end-to-end visibility into revenue trends, product performance, payment mode adoption, and regional state-level breakdown across 191 granular order records.

Dataset Overview

Attribute	Detail
Source	Kaggle — Ecommerce sales dataset
Records	191 rows
Columns	13 (Order ID, Order Date, Customer Name, Category, Payment Mode, Ship Mode, State, City, Quantity, Sales, Profit, and more)
Time Period	January – December 2023
Granularity	One row per order transaction

Key Fields Used

ORDER DATE	CATEGORY	PAYMENT MODE	STATE / CITY	QUANTITY	SALES	PROFIT
CUSTOMER NAME						

Data Preparation

Validated data types, cleaned missing or inconsistent values, created a Calendar table, established relationships, and built DAX measures for reporting. In more detail:

- 1 Changed data types to ensure numerical and date accuracy
- 2 Validated and cleaned the dataset for consistency
- 3 Built a dedicated Calendar Table using the `CALENDAR()` DAX function
- 4 Created Year, Month, Month Number, and Quarter columns for time-based analysis
- 5 Marked the Calendar table as the official Date Table
- 6 Established a relationship between the Sales and Calendar tables using Order Date
- 7 Sorted Month by Month Number to display visuals in chronological order

2 Report Structure

The report is built across two pages, designed to move from a project overview into a full interactive analysis view.



Page 1 — Cover

Project title, prepared-by details, a short description of the report's purpose, and a "Dashboard Includes" panel listing every analysis area.



Page 2 — Executive Performance Report

The central analytics hub: KPI cards, four global slicers (Payment Mode, Category, State, Order Month), a Geographic Map Visual (Total Sales by State), and three charts — Sales by Category, Sales by Payment Mode, and Sales Trend by Month.

3 Interactive Features



Geographic Map Visual (Total Sales by State)

An interactive map displaying sales distribution across states, with bubble scaling used to visually indicate revenue concentration and highlight core regional markets.



Custom Calendar Table (DAX)

Built using the DAX CALENDAR function with Year, Month, Month Number, and Quarter columns — marked as the official date table and related to Sales on Order Date, then sorted chronologically.

DAX Measures Created

- **Total Sales** = `SUM(Sales[Sales])`
- **Total Profit** = `SUM(Sales[Profit])`
- **Total Orders** = `DISTINCTCOUNT(Sales[Order ID])`
- **Total Customers** = `DISTINCTCOUNT(Sales[Customer Name])`
- **Total Quantity** = `SUM(Sales[Quantity])`

3 Interactive Features (continued)



Tooltip on Cover Page

A custom tooltip surfaces extra context on hover, giving viewers a quick orientation before they move into the dashboard.



Page Navigation

Custom navigation lets users move between the Cover page and the Executive Performance Report without relying on page tabs.



Slicers

Payment Mode, Category, State, and Order Month slicers let viewers filter every visual on the page simultaneously.

4 Key Metrics at a Glance (Jan – Dec 2023)

191

TOTAL CUSTOMERS

163

TOTAL ORDERS

840

TOTAL QUANTITY

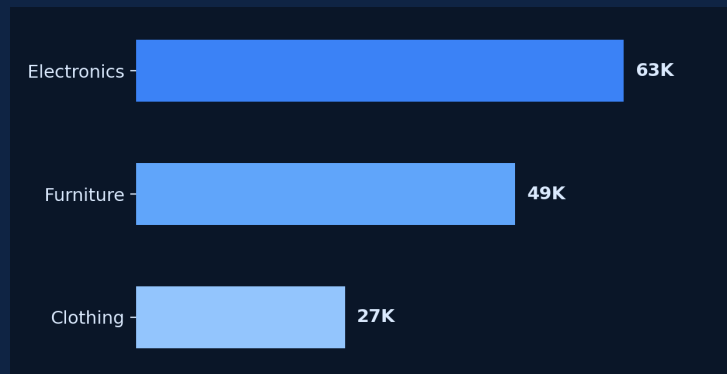
\$139.12K

TOTAL SALES

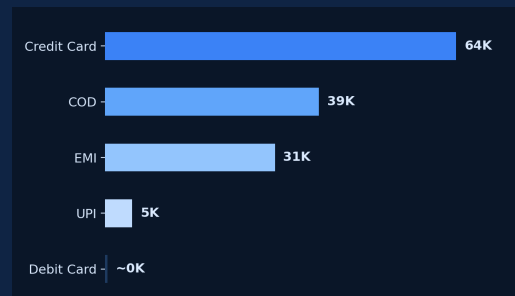
\$15.24K

TOTAL PROFIT

Sales by Category

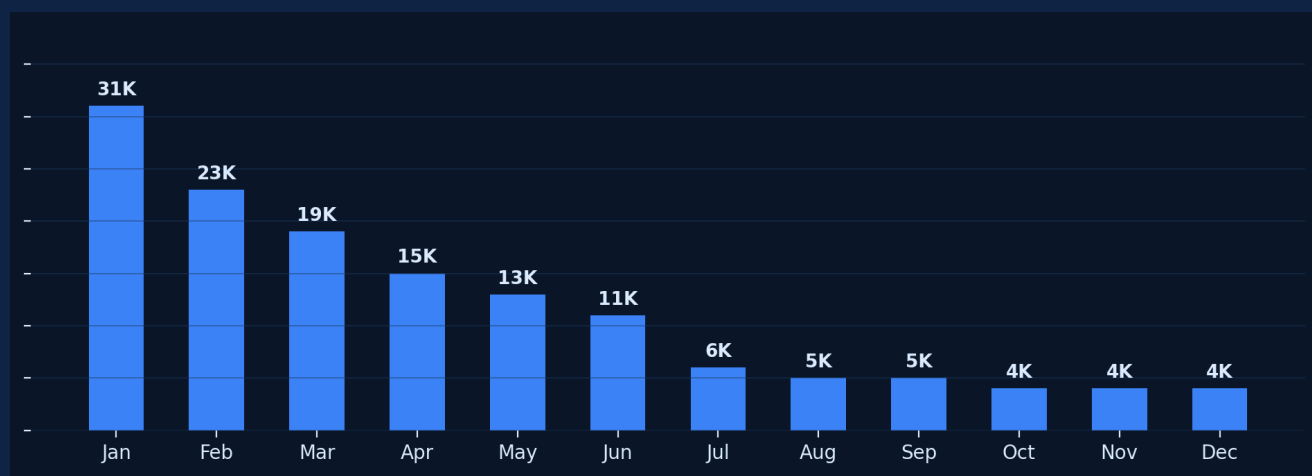


Sales by Payment Mode



Sales Trend by Month

Column chart displaying monthly sales performance.



5 Business Insights

Regional concentration (map visual analysis).

The Total Sales by State map reveals strong demand clusters centered in key metropolitan states, enabling targeted inventory placement and state-specific promotional strategies.

Electronics drives bulk revenue.

Electronics is the top-performing category at \$63K (45%) in sales, outperforming Furniture (\$49K / 35%) and Clothing (\$27K / 19%).

Card-based payments lead transactions.

Credit Card is the leading payment mode (\$64K), followed by Cash on Delivery (\$39K) and EMI (\$31K). Digital-first modes like UPI (\$5K) remain underutilized — an opportunity for checkout conversion incentives.

Front-loaded monthly trend.

Sales performance in 2023 was highest during Q1, peaking in January at \$31K, followed by steadier baseline volumes across H2 2023.

6 Conclusion

This Power BI dashboard transforms raw e-commerce order records into an executive decision-making tool. By coupling comprehensive DAX modeling with geographical state mapping and dynamic filtering, the dashboard provides a practical template for retail performance monitoring.

Tools Used

[Power BI Desktop](#)
[Power Query](#)
[DAX Modeling](#)
[Geospatial Mapping](#)
[Interactive Slicers](#)